TEST 7 INNOVATOR
MATH MODULE 1

1. Which expression is equivalent to $(5x^2 + 5x) - (4x^2 - 2x)$?

NEXT QUESTION

- A $-9x^2 + 7x$
- B $-9x^2 + 3x$
- C $x^2 + 3x$
- D $x^2 + 7x$

2. If $(x - 1)(x + 3) = k + 2$ and $x = 2$, what is the value of k?

PREV. QUESTION NEXT QUESTION

- A 0
- B 2
- C 3
- D 5

3. If $\frac{28}{9}z = \frac{35}{3}$, what is the value of z?

PREV. QUESTION NEXT QUESTION

- A $\frac{4}{15}$
- B $\frac{4}{5}$
- C $\frac{5}{4}$
- D $\frac{15}{4}$

4. If $k > 0$ and $k^4 + 5 = 21$, what is the value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

5. $$-4(9 - 12x) = 6(8x + 6)$$

How many solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

A Exactly one

B Exactly two

C Infinitely many

D Zero

6. Right triangles JKL and MNO are similar such that J, K, and L correspond to M, N, and O, respectively. If angle L is a right angle and the measure of angle J is 32°, what is the measure of angle N, in degrees?

PREV. QUESTION | NEXT QUESTION

A 32

B 58

C 90

D 122

7. If $w = 12\sqrt{2}$ and $\frac{1}{4}w = \sqrt{2x}$, what is the value of x?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

8. The function f is defined by $f(x) = 9x + 36$. The graph of $y = f(x)$ in the xy-plane has an x-intercept of $(h, 0)$ and a y-intercept of $(0, k)$, where h and k are constants. What is the value of $h + k$?

PREV. QUESTION | NEXT QUESTION

A) 4

B) 32

C) 36

D) 40

9. A tree with an initial height of 1 meter grows at an average rate of 1.2 meters per year. At that rate, how many years will it take for the tree to reach a height of 19 meters?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

10. If $14p - 22 \geq -10$, what is the maximum possible value of $11 - 7p$?

PREV. QUESTION | NEXT QUESTION

A. -5

B. $\frac{6}{7}$

C. $\frac{16}{7}$

D. 5

11. A soccer coach wants to purchase new uniforms for his 22 players. The athletic department's budget allows for a maximum of $\$2{,}800$ to be spent on these uniforms, which includes a 6% sales tax. Which of the following is the maximum amount of money per uniform, before sales tax, that the coach can spend?

PREV. QUESTION | NEXT QUESTION

A. $\$113.27$

B. $\$120.06$

C. $\$127.27$

D. $\$135.39$

12. $$A = 14.6 + 2.35d$$
$$B = 6.9 + 3.45d$$

In the given system of equations, A and B represent the heights, in centimeters, of two plants d days after a starting date. For what value of d will the plants have equal heights?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13. What is the diameter of a circle with the equation $(x-3)^2+(y+5)^2=64$?

PREV. QUESTION | NEXT QUESTION

- A 8
- B 16
- C 64
- D 128

14. $$y=-x^2+12x+10$$

The given equation relates the variables x and y. For what value of x does the value of y reach its maximum?

PREV. QUESTION | NEXT QUESTION

- A -6
- B 5
- C 6
- D 10

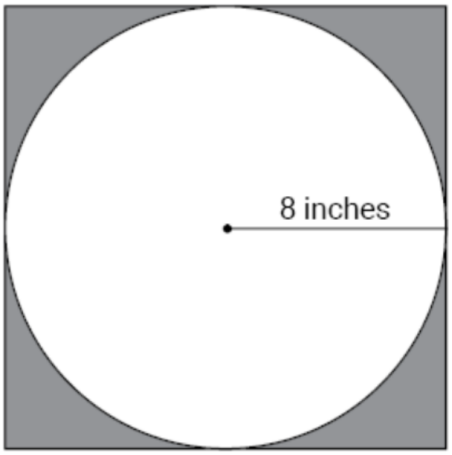
15. $$x(2x-6)+18=x(5x+9)$$

What is the sum of the solutions to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

16.



A circle is inscribed in a square, as shown. What is the area of the shaded region, in square inches?

PREV. QUESTION | NEXT QUESTION

A) $256 - 64\pi$

B) $256 - 32\pi$

C) $256 - 16\pi$

D) $256 - 8\pi$

COME BACK LATER

17.

	Blood Type				
Rhesus Factor	A	B	AB	O	Total
+	34	46	18	42	140
–	21	29	28	22	100
Total	55	75	46	64	240

Humans can have one of four blood types: A, B, AB, and O. Each blood type also has an rhesus factor (Rh) that is either positive or negative. A hospital held a blood drive, and the table above shows the distribution of blood types and Rh factors for the units of blood collected. What is the probability that a randomly selected unit is O negative (–)?

PREV. QUESTION | NEXT QUESTION

A) $\frac{11}{120}$

B) $\frac{11}{50}$

C) $\frac{11}{32}$

D) $\frac{11}{21}$

18.

x	$f(x)$
−3	9
−1	$\frac{19}{3}$
2	$\frac{7}{3}$
12	−11

The table shows select values for the function $f(x)$. What is the equation for the function $f(x)$?

PREV. QUESTION NEXT QUESTION

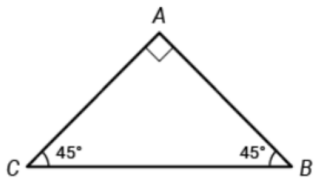
A $f(x) = -3x + 9$

B $f(x) = -\frac{4}{3}x + \frac{13}{3}$

C $f(x) = -\frac{4}{3}x + 5$

D $f(x) = \frac{4}{3}x + 5$

19.



Note: Figure not drawn to scale

In the figure shown, the length of BC is 6 centimeters. What is the length of AB, in centimeters?

PREV. QUESTION NEXT QUESTION

A 3

B $3\sqrt{2}$

C 6

D $6\sqrt{2}$

20. $$-12x^2 - bx - 12 = 0$$

In the given equation, b is a constant. For which of the following values of b will the equation have no real solutions?

PREV. QUESTION | NEXT QUESTION

A -25

B -24

C 12

D 24

21. Data set X consists of 10 positive integers less than 75. The list shown gives 9 of the integers from data set X.

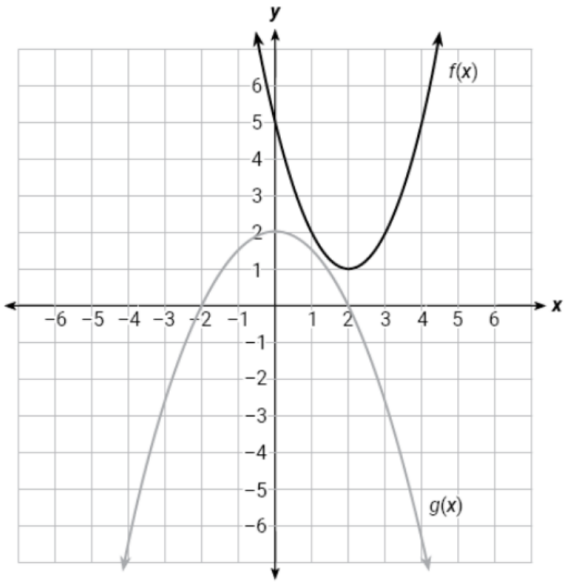
$$60, 64, 56, 68, 58, 60, 56, 58, 69$$

The total sum of the 10 integers is 620. If the mean of data set X is an integer that is greater than 61, what is the value of the largest integer from data set X?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22.



The functions $f(x)$ and $g(x)$ are graphed on the xy–plane shown. If $g(2) = q$, what is the value of $f(q)$?

PREV. QUESTION

A 0

B 1

C 2

D 5

COME BACK LATER

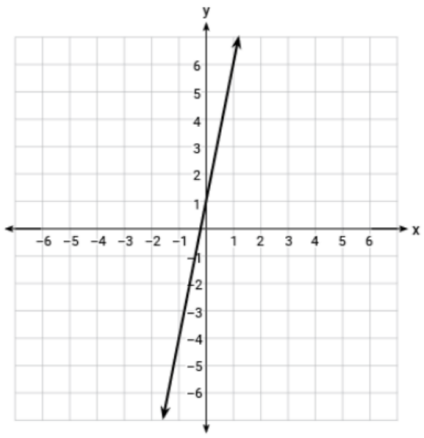
MATH MODULE 2 (HARDER)

1. If $6n + 40 = 3(2n - 20) + 2n$, what is the value of $2n$?

NEXT QUESTION

COME BACK LATER

2.



The graph of line q is shown above. Line r is perpendicular to line q. What is the slope of line r?

PREV. QUESTION

NEXT QUESTION

A -5

B $-\frac{1}{5}$

C $\frac{1}{5}$

D 5

COME BACK LATER

3. $$nA = 360$$

The above equation shows the relationship between the measure A, in degrees, of an exterior angle of a regular polygon and n the number of sides of the polygon. If a regular polygon's exterior angle is greater than 80 degrees, what is the greatest number of sides it can have?

PREV. QUESTION

NEXT QUESTION

A 4

B 5

C 6

D 8

4. For the linear function $g(x)$, $g(2.6) = 11$ and $g(5) = 3.8$. When graphed on the coordinate plane, the function $g(x)$ is perpendicular to the graph of the linear function $h(x) = mx$. What is the value of m?

PREV. QUESTION | NEXT QUESTION

- A -3
- B $-\frac{1}{3}$
- C $\frac{1}{3}$
- D 3

5. A right triangle has legs with lengths of 15 centimeters and 25 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $5\sqrt{n}$, where n is an integer, what is the value of n?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6. Harold prepared a sample of bacteria. A strain of virus killed 30 percent of the bacteria, but within a week, the surviving population had grown 47 percent. Which of the following represents the value of the original population in terms of the new population p?

PREV. QUESTION | NEXT QUESTION

- A $\frac{p}{1.17}$
- B $1.17p$
- C $\frac{p}{(0.70)(1.47)}$
- D $(0.70)(1.47)p$

7. The population of fruit flies in a laboratory study triples every 7 days. The population at the beginning of the study numbered 200. Which equation represents the population of fruit flies, y, in the study t weeks after the beginning of the study?

PREV. QUESTION | NEXT QUESTION

- A $y = 200(3)^t$
- B $y = 3(200)^t$
- C $y = 200(3)^{\frac{t}{7}}$
- D $y = 200^{3t}$

8. $$24x + 24y = -8y + \frac{2}{3}$$
$$-ky = 3x - \frac{1}{12}$$

In the given system of equations, k is a constant. If the system has infinite solutions, what is the value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

9. As a probe descends to the Kuril Trench in the Pacific Ocean, it records the amount of dissolved oxygen in the water in parts per million (ppm). At depths below $2{,}000$ meters, the dissolved oxygen drops at a constant rate of d ppm for every change in depth of 100 meters until it reaches zero. At a depth of $2{,}530$ meters, the dissolved oxygen is measured to be 205 ppm. At a depth of $3{,}030$ meters, the dissolved oxygen is measured to be 65 ppm. What is the value of d?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

10. Mrs. Brennan wants to buy x computers for her company and has a budget of B dollars. If she buys x computers today, they will cost $\$1{,}000$ each, and she will need to raise an additional $\$2{,}000$. If she waits two months, the computers will cost $\$900$ each, and she will have $\$400$ left over. What is the value of B?

PREV. QUESTION | NEXT QUESTION

- A $14{,}800$
- B $21{,}600$
- C $22{,}000$
- D $24{,}000$

11. $$f(x) = x(x - 14) + 70 - k$$

In the given function, k is an integer constant. For which of the following values of k will the graph of the function have no x-intercepts?

PREV. QUESTION | NEXT QUESTION

- A 20
- B 21
- C 25
- D 30

12. A professional football team practices 6 days a week in the preseason. A water boy for the team knows that the players drink about 200 gallons of water each day of preseason practice. If the water tank in the utility room pours water at a rate of 9 gallons per minute, how many hours a week does the water boy spend filling water containers?

PREV. QUESTION | NEXT QUESTION

A $\frac{10}{27}$

B $\frac{20}{9}$

C $\frac{100}{27}$

D $\frac{400}{3}$

13. The graph of $x^2 + 3x + y^2 - y = \frac{13}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

14. In the xy-plane, a parabola has a vertex of $(3, -10)$ and intersects the x-axis at two points. If the equation of the parabola is in the form $y = ax^2 + bx + c$, where a, b, and c are constants, which of the following could be the value of $a + b + c$?

PREV. QUESTION | NEXT QUESTION

A -6

B -10

C -12

D -14

15. $$3(ax + b) = \frac{21}{10}x - \frac{48}{11}$$

In the given equation, a and b are constants and $b > 0$. The equation has no solution. What is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

16. Which expression is equivalent to $\frac{1}{\frac{1}{x+5} + \frac{1}{x-2}}$?

PREV. QUESTION | NEXT QUESTION

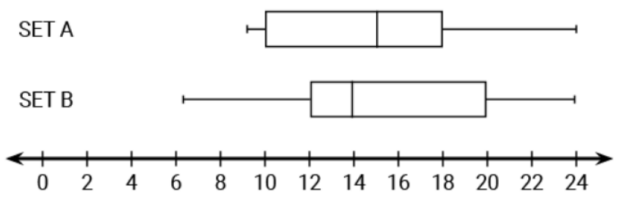
A $\frac{2}{2x+3}$

B $\frac{2x+3}{2}$

C $\frac{2x+3}{x^2+3x-10}$

D $\frac{x^2+3x-10}{2x+3}$

17.

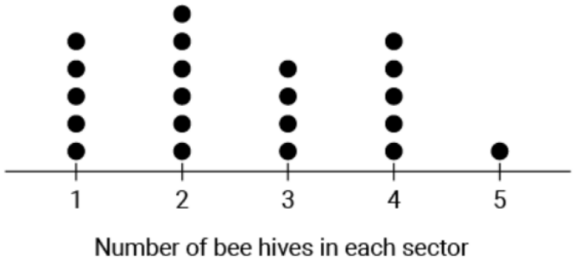


The box plot summarizes the values in two different data sets. Based on the box plot, which of the following must be true?

PREV. QUESTION NEXT QUESTION

- A The mean of set A is greater than the mean of set B.
- B The range of set A is greater than the range of set B.
- C The median of set A is greater than the median of set B.
- D The standard deviation of set A is greater than the standard deviation of set B.

18.



Number of bee hives in each sector

Researchers studying the bee population in Epsilon Park divided the park into 21 sections of identical areas and recorded how many bee hives were found in each sector. The results are shown in the dot plot. If the sector that had 5 bee hives was removed from the data, which of the following statements is true?

PREV. QUESTION NEXT QUESTION

- A The median and mean will decrease.
- B The median and mean will stay the same.
- C The median will decrease and the mean will stay the same.
- D The median will stay the same and the mean will decrease.

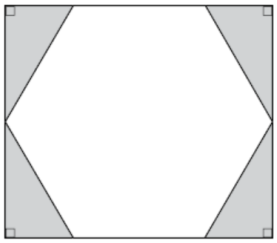
19. $0.6x^2 - 1.5x + 2.01 = 0$

How many distinct real solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

- A Zero
- B Exactly one
- C Exactly two
- D Infinitely many

20. A regular hexagon is inscribed in a rectangle as shown.



The perimeter of the hexagon is $6a$ units. What is the area of the shaded region, in square units?

PREV. QUESTION | NEXT QUESTION

- A $\frac{\sqrt{3}}{8}a^2$
- B $\frac{\sqrt{3}}{2}a^2$
- C $\frac{3\sqrt{3}}{2}a^2$
- D $2\sqrt{3}a^2$

21. The function $g(x)$ is defined as $g(x) = (x^2 + bx + c)(x + a)$, where a, b, and c are constants. There are two distinct values such that $g(x) = 0$. If $g(x)$ can also be written in the form of $\frac{1}{m}(mx^2 + mbx + b^2)(x + a)$, what is the value of m?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22. The function f is defined by the equation $f(x) = \frac{x}{a-9}$, where a is a positive constant and $a \neq 9$. If the value of the product of $f(2a)$ and $f(18a)$ is equal to $\frac{9}{100}$, what is the value of a?

PREV. QUESTION

COME BACK LATER